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Original Article

Safety and effectiveness of lecanemab in Chinese patients with early Alzheimer's disease: Evidence from a multidimensional real-world study

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Conflicts of interest

None.

Abstract

Introduction: Lecanemab has shown promise in treating early Alzheimer's disease (AD), but its safety and efficacy in Chinese populations remain unexplored. This study aimed to evaluate the safety and 6-month clinical outcomes of lecanemab in Chinese patients with mild cognitive impairment (MCI) or mild AD.

Methods: In this single-arm, real-world study, participants with MCI due to AD or mild AD received biweekly intravenous lecanemab (10 mg/kg). Safety assessments included monitoring for amyloid-related imaging abnormalities (ARIA) and other adverse events. Clinical and biomarker changes from baseline to 6 months were evaluated using cognitive scales (mini-mental state examination [MMSE], montreal cognitive assessment [MoCA], clinical dementia rating–sum of boxes [CDR-SB]), plasma biomarker analysis, and advanced neuroimaging.

Results: A total of 64 patients were enrolled in this ongoing real-world study. Safety analysis revealed predominantly mild adverse events, with infusion-related reactions (20.3%, 13/64) being the most common. Of these, 69.2% (9/13) occurred during the initial infusion and 84.6% (11/13) did not recur. ARIA-H (microhemorrhages/superficial siderosis) and ARIA-E (edema/effusion) were observed in 9.4% (6/64) and 3.1% (2/64) of participants, respectively, with only two symptomatic cases (one ARIA-E presenting with headache and one ARIA-H with visual disturbances). After 6 months of treatment, cognitive scores remained stable compared to baseline (MMSE: 22.33 ± 5.58 *vs*. 21.27 ± 4.30, P =

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0.733; MoCA: 16.38 ± 6.67 *vs*. 15.90 ± 4.78, P = 0.785; CDR-SB: 2.30 ± 1.65 *vs*. 3.16 ± 1.72, P = 0.357), while significantly increasing plasma amyloid-β 42 (Aβ42) (+21.42%) and Aβ40 (+23.53%) levels compared to baseline.

Conclusions: Lecanemab demonstrated a favorable safety profile in Chinese patients with early AD. Cognitive stability and biomarker changes over 6 months suggest potential efficacy, though high dropout rates and absence of a control group warrant cautious interpretation. These findings provide preliminary real-world evidence for lecanemab's use in China, supporting further investigation in larger controlled studies.

Registration: ClinicalTrials.gov, NCT07034222.

Keywords: Alzheimer's disease; Lecanemab; Chinese; Safety; Cognitive function; Plasma biomarkers; Neuroimaging

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Background

Alzheimer's disease (AD), the most prevalent neurodegenerative disorder associated with dementia, accounts for 60–70% of all dementia cases globally. AD begins with memory decline and progresses to severe cognitive impairments, ultimately rendering individuals bed-bound and in need of constant care. Major pathological hallmarks of AD include amyloid plaques from amyloid-β (Aβ) peptide deposition and neurofibrillary tangles from abnormal tau protein aggregation in the brain.[1]

The amyloid cascade hypothesis suggests that Aβ accumulation in the brain is central to AD development. Aβ aggregation, especially toxic forms like Aβ42, triggers pathological processes including synaptic dysfunction, tau hyperphosphorylation, neurofibrillary tangles, and neuronal cell death.[2,3] Although the Aβ hypothesis has driven AD research for decades, ongoing debates persist regarding its central role in pathogenesis.[4,5] Nevertheless, targeting Aβ remains a key therapeutic

strategy, as evidenced by the development of anti-Aβ therapies, such as lecanemab, aducanumab, and donanemab.[6–8]

Lecanemab, a humanized immunoglobulin G1 (IgG1) monoclonal antibody targeting Aβ protofibrils,[9] has shown significant promise in the treatment of AD in the phase 3 Clarity AD trial.[6] Over 18 months of treatment, it significantly slowed global cognitive decline compared to placebo (clinical dementia rating–sum of boxes [CDR-SB] difference: –0.45; 95% confidence interval [CI], –0.67 to –0.23, P <0.001).[6] The safety profile of lecanemab, particularly the incidence of amyloid-related imaging abnormalities (ARIA), is closely linked to Apolipoprotein E (APOE) ε4 genotype. The rate of APOE ε4 homozygotes was 15.4% in the Clarity AD study[6] and 14.1% in Northern European,[10] which differ from previously reported rates in Chinese AD populations (5.72–16.72%).[11] This genotypic discrepancy may have important implications for the drug's safety profile in Chinese patients. Despite lecanemab's therapeutic promise, post-marketing surveillance has raised safety concerns, including ARIA.[12] Furthermore, despite evidence showing a reduction in Aβ levels and slowing of cognitive decline, long-term efficacy and impact on disease progression remain underexplored. Therefore, careful monitoring and risk assessment, especially in comorbid patients and those carrying the ApoE ε4 gene, is crucial,[13] and the benefits observed in clinical trials need to be validated in real-world settings.

The AD field is increasingly calling for more real-world data on the clinical application of lecanemab. As the most populous country in the world, China had an estimated 13,143,950 cases of AD and other dementias, representing approximately 25.5% of the global disease burden;[14] however, existing clinical trials of lecanemab have primarily enrolled Western populations.[12,15] While the pivotal Phase 3 clinical trial for lecanemab included sites in China, contributing to its global dataset,

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dedicated studies focusing exclusively on the efficacy and safety of lecanemab in Chinese populations remain limited. Emerging evidence from Chinese cohorts is beginning to complement these findings.[16,17] Genetic factors, such as APOE ε4 allele, may influence disease progression, drug response, and safety profiles.[18,19] In addition, environmental factors, including lifestyle factors, also modulate disease progression.[20] Importantly, AD progression or its measurement may differ across geographic regions.[21] Given the distinct genetic background, unique lifestyle patterns, and growing disease burden in China, there is a pressing need to conduct population-specific research on lecanemab.

Under the support of China's "Pioneering and Pilot" policy, Hainan Branch, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, has conducted the first real-world study of lecanemab in Chinese patients with mild cognitive impairment (MCI) and mild AD. This study adopts a multimodal assessment framework encompassing safety monitoring, cognitive outcomes, plasma biomarkers, and neuroimaging metrics. These data aim to provide preliminary evidence for localized clinical decision-making and to inform long-term studies tailored to the Chinese population.

Methods

Protocol approval and informed consent

The study was conducted at Hainan Branch, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine in accordance with the ethical standards of the 1964 *Declaration of Helsinki*. Protocol and informed consents were approved by the Ethics Committee of Hainan Branch, Ruijin Hospital, Shanghai Jiao Tong University School of Medicine (No. 2024-021). All participants and their study partners provided written informed consent.

Eligibility criteria

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The study enrolled participants 50 to 90 years of age, with a clinical diagnosis of either MCI due to AD or mild AD based on National Institute on Aging–Alzheimer's Association criteria.[22,23] Confirmation of Aβ pathology was required via Aβ-positron emission tomography (PET) imaging or cerebrospinal fluid (CSF) Aβ42/Aβ40 ratio analysis (results obtained within 6 months). Cognitive function was assessed using standardized scales completed within 3 months: Clinical Dementia Rating (CDR) global score of 0.5–1 and/or mini-mental state examination (MMSE) score of 20–30. Availability of APOE genotype results was mandatory. The methodology for ApoE genotyping is shown in Supplementary Materials, http://links.lww.com/CM9/C677. Magnetic resonance imaging (MRI) / susceptibility-weighted imaging (SWI) eligibility was confirmed by the absence of exclusionary findings. Routine laboratory tests, thyroid function, vitamin B12, and coagulation tests were normal or clinically insignificant; syphilis, human immunodeficiency virus (HIV), and other cognition-related infections were negative.

Patients were excluded if MRI/SWI showed ≥4 microhemorrhages (≤10 mm), macrohemorrhages (>10 mm), cortical superficial siderosis, vasogenic edema, brain contusion/encephalomalacia/vascular malformations/infectious lesions, multiple lacunar infarcts/strokes, severe small vessel disease, severe white matter lesions (Fazekas ≥3), or space-occupying lesions (except asymptomatic meningiomas/arachnoid cysts <1 cm).

Study design

This ongoing prospective study, registered on ClinicalTrials.gov (Identifier: NCT07034222), evaluates lecanemab treatment (10 mg/kg biweekly intravenous infusion) in patients with early AD. As a prospective, observational, real-world study conducted under China's "Pioneering and Pilot" policy, it was designed to characterize the initial clinical experience and safety profile of lecanemab in a Chinese

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clinical setting. Patient enrollment and baseline assessments commenced in November 2023. In line with the descriptive and hypothesis-generating nature of such studies, a formal *a priori* sample size calculation was not performed. Instead, the study aimed to enroll all eligible patients who initiated treatment at our center during the designated recruitment period. According to the registered protocol, all participants undergo comprehensive assessments at baseline, 3, 6, and 12 months, including standardized cognitive assessments, plasma AD biomarker testing, and advanced neuroimaging. A subset of patients voluntarily undergoes additional Aβ-PET imaging at baseline and follow-up. The current interim analysis reports safety and efficacy outcomes for the first 64 enrolled patients, focusing on data from baseline, 3 months, and 6 months, as only 4 patients had completed the 12-month follow-up at the time of data cutoff for this analysis. Aβ-PET data from these 4 patients at 12 months are also presented. Follow-up methods included both in-person visits with clinical and MRI assessments, as well as remote monitoring through telephone interviews and centralized review of transmitted MRI reports and images (via QR code sharing) for non-local participants. Follow-up is ongoing to collect additional longitudinal data.

Safety analyses

Safety was assessed through adverse event monitoring, vital signs, and physical examinations. The occurrence of ARIA was monitored throughout the study via MRI scans before the 5th, 7th, 14th, and 26th doses of treatment, corresponding to weeks 9, 13, 27, and 51, respectively. ARIA was assessed for radiographic severity, the presence of symptoms, and the clinical severity of symptomatic ARIA. Safety assessments were conducted in the safety analysis set.

Clinical assessment

Comprehensive evaluations were conducted of the patients' global cognitive function using the

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MMSE, montreal cognitive assessment (MoCA), and CDR-SB. The MMSE has a total score ranging from 0 to 30, with higher scores indicating better cognitive function. Similarly, the MoCA also ranges from 0 to 30, with higher scores reflecting better cognitive performance. CDR-SB scores range from 0 to 18, with higher scores indicating greater impairment. Daily activities were evaluated using the Functional Activities Questionnaire (FAQ). The burden of caregivers was evaluated using the Zarit Burden Interview (ZBI).

Plasma biomarker analyses

Biomarker sample analysis was conducted at Frontage Laboratories (Shanghai, China), focusing on phosphorylated tau 181 (p-tau181), phosphorylated tau 217 (p-tau217), Aβ42, Aβ40, glial fibrillary acidic protein (GFAP), and neurofilament light chain (NfL). Measurements were performed using the Quanterix Simoa® HD-X ultrasensitive single-molecule analyzer with p-Tau 181 Advantage PLUS, Simoa ALZpath p-Tau 217 Advantage PLUS, and Simoa Neuro 4-Plex E Advantage assay kits. Serial plasma samples were collected at baseline, 3 months, and 6 months. Samples were batch-analyzed to reduce assay variability and ensure data consistency. The detailed methodology and assay performance characteristics, such as the limit of detection, limit of quantification, precision (coefficient of variation, CV%), and linear range for each assay (p-tau181, p-tau217, Aβ40, Aβ42, GFAP, and NfL), are provided in Supplementary Materials, http://links.lww.com/CM9/C677.

MRI acquisition and analysis

Acquisition of multimodal MRI

MRI data were acquired on a 3T scanner with a 64-channel head coil as in our previous RJNB-Dementia protocol.[24] Detailed acquisition parameters and scanner settings are provided in Supplementary Materials, http://links.lww.com/CM9/C677.

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Rs-fMRI analysis

Resting-state functional magnetic resonance imaging (Rs-fMRI) is a technique that assesses brain connectivity by measuring spontaneous neural activity while a subject is not engaged in any specific task. Rs-fMRI data were preprocessed using the Data Processing Assistant for Resting-State fMRI (DPARSF) toolbox in MATLAB, including slice timing correction, head motion realignment, spatial normalization to MNI152 space, and smoothing with a 4–6 mm full-width at half-maximum (FWHM) Gaussian kernel. Detailed descriptions of preprocessing steps, parameter settings, and analytic procedures are available in the Supplementary Materials, http://links.lww.com/CM9/C677.

Diffusion MRI analysis

Susceptibility-induced distortions were corrected using the top-up method, followed by head motion and eddy current correction using the Eddy tool, which is part of FSL (FMRIB Software Library, a comprehensive suite of tools for neuroimage analysis). Diffusion tensors were fitted at each voxel using FSL's DTIFIT tool (Diffusion Tensor Imaging Fit Tool), which performs voxel-wise diffusion tensor modeling to generate maps for fractional anisotropy (FA), mean diffusivity (MD), radial diffusivity (RD), and axial diffusivity. For spatial normalization, FA maps were non-linearly registered to the MNI152 FA template, with co-registration accuracy visually inspected to ensure proper alignment. White matter tracts of interest were defined using the JHU-ICBM-tracts-maxprob-atlas, and masks were applied to the FA, MD, RD, and axial diffusivity maps to extract mean values for each tract. These mean values were then subjected to statistical analysis.

PET scan

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Static PET images were acquired 40–60 min post-injection. Image reconstruction was performed using the ordered subset expectation maximization algorithm, followed by smoothing with a 2-mm FWHM Gaussian kernel. Standardized uptake value ratio (SUVR) maps for [^{18}F] florbetapir were calculated using the entire cerebellum as the reference region.

Statistical analysis

Analysis sets

As of the data cutoff, all 64 enrolled patients had reached the 6-month primary endpoint of the current study. The safety analysis set included all 64 patients who received at least one dose of lecanemab and had at least one ARIA evaluation at week 9. The efficacy analysis set was defined as the full analysis set, comprising all patients in the safety analysis set. Due to a substantial amount of missing data resulting from participant dropout during the follow-up period, the analysis of longitudinal efficacy endpoints was performed using a generalized linear mixed-effects models (GLMMs). This model incorporated all available observed data from baseline, 3-month, and 6-month assessments to provide valid estimates under the missing-at-random (MAR) assumption.

Statistical methods

We presented normally distributed continuous variables as mean ± standard deviation, non-normally distributed continuous variables as median with interquartile range (IQR), and categorical variables as percentages. For the longitudinal analyses, we used GLMMs under the MAR assumption, with time specified as a fixed effect and a random intercept included for each subject to account for within-subject correlations due to repeated measures. These models were applied to evaluate changes in clinical scale scores, plasma biomarkers, and neuroimaging parameters during treatment. Based on prior literature, statistical principles, and clinical expertise, we adjusted for

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baseline age, sex, years of education, ApoE ε4 carrier status, and baseline MMSE score when analyzing plasma and neuroimaging data. For clinical scale data, the models were adjusted for baseline age, sex, years of education, and ApoE ε4 carrier status. In addition, we included the interaction term between ApoE ε4 carrier status and time to explore whether ApoE ε4 modulated longitudinal changes in treatment response. Normality of outcome variables was examined using the Shapiro–Wilk test. Non-normally distributed variables were log-transformed to reduce skewness. To control the false discovery rate, statistical results from the GLMM analyses were corrected using the Benjamini–Hochberg (BH) procedure. Under the MAR assumption, GLMMs can incorporate all available repeated-measures data to provide efficient inferences. Given the substantial attrition in scale data, we additionally performed multiple imputation with 5 imputations and 30 iterations as a sensitivity analysis. The imputed datasets were used for subsequent GLMM analyses to supplement and validate the robustness of the primary findings. To assess whether the data missing due to attrition introduced selection bias, we compared baseline characteristics between patients who completed the 6-month follow-up and dropouts. Student's *t*-test was used for normally distributed continuous variables, the Mann–Whitney *U* test for non-normally distributed continuous variables, and Fisher's exact test for categorical variables. All analyses were performed in R software 4.3.3 (R Foundation for Statistical Computing, Vienna, Austria), and statistical significance was defined as $P < 0.05$.

Results

Demographics and clinical characteristics of enrolled participants

Patient enrollment and baseline assessments commenced in November 2023.This ongoing real-world study has enrolled 64 patients to date, with plans for continued recruitment and extended follow-up. A detailed flow diagram of patient enrollment and follow-up has been provided as Figure 1.

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The study population had a median (IQR) age of 68.0 (61.0–74.0) years with equal gender distribution (50% female). Based on the global CDR score, 89.1% (57/64) of the participants were classified as having MCI, while 10.9% (7/64) met criteria for mild AD. The APOE ε4 allele was present in 67.2% (43/64) of participants (62.5% [40/64] heterozygous; 4.7% [3/64] homozygous). Baseline antiplatelet use, exclusively with aspirin, was reported in 7.8% (5/64) of participants. Complete demographic and clinical characteristics are presented in Table 1.

Safety

Of the enrolled cohort, seven patients discontinued lecanemab: one after 15 doses due to negative follow-up Aβ-PET results, another after 10 doses due to self-reported lack of improvement, and five due to treatment-emergent adverse events (TEAEs). All patients underwent systematic AE monitoring. Safety analysis revealed that 56.3% (36/64) of participants experienced at least one adverse event. The most frequently reported TEAEs included infusion-related reactions (20.3%, 13/64), headache (12.5%, 8/64), ARIA-H (9.4%, 6/64), dizziness (7.8%, 5/64), and decreased appetite (6.3%, 4/64). Five participants (7.8%, 5/64) discontinued due to TEAEs, including one case of erectile dysfunction with concurrent dizziness, one case of hydrocephalus, one case of atrial fibrillation complicated by heart failure, and two cases of ARIA-H, presenting as cerebral microhemorrhages.

Infusion-related reactions primarily manifested as flu-like symptoms (fever, chills, myalgia, and arthralgia; 9.4% [6/64]), hypotension (4.7% [3/64]), oxygen desaturation (3.1% [2/64]), nausea (1.6% [1/64]), and hypertension (1.6% [1/64]). These reactions were predominantly mild, with 69.2% (9/13) occurring during the initial infusion. No prophylactic medications (e.g., antihistamines, hydrocortisone) were administered before subsequent infusions. Notably, 84.6% (11/13) of patients did not experience recurrent infusion reactions. Complete TEAE data are presented in Table 2.

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All participants completed at least one post-baseline MRI scan to monitor for ARIA-E and ARIA-H. Post-baseline MRI revealed ARIA-H in 9.4% (6/64) of participants and ARIA with edema (ARIA-E) in 3.1% (2/64). Radiographic findings were predominantly mild, with symptomatic presentations including one case of ARIA-E characterized by mild headache and one case of ARIA-H manifesting as transient visual disturbances (blue dots, clouding, and perceived ceiling movement). Among patients experiencing ARIA-E or ARIA-H, five cases were APOE ε4 heterozygous carriers and three cases were non-carriers. Four patients remained stable based on MRI surveillance, including three at 12 months and one at 8 months. Notably, none of these patients were taking antiplatelet or anticoagulant drugs at baseline or during the treatment period. Treatment discontinuation occurred in two cases: one patient withdrew after the 13th dose following a diagnosis of hydrocephalus at a non-study center, and another discontinued therapy due to concerns regarding mild ARIA-H. One patient had a complete remission of ARIA-E. One patient developed severe ARIA-H at 12 months, although the ARIA-E had resolved completely by 12 months. Detailed radiological and clinical severity assessments, along with APOE genotypes, are summarized in Supplementary Table 1, http://links.lww.com/CM9/C677.

We further evaluated the potential influence of ApoE genotype on ARIA risk. Participants were categorized as ε4 non-carriers (n = 21), ε4 heterozygotes (n = 40), or ε4 homozygotes (n = 3). Among the eight participants who developed ARIA, the incidence was 14.3% (3/21) in non-carriers and 12.5% (5/40) in heterozygotes. No ARIA cases were observed in the three ε4 homozygotes. Fisher's exact test revealed no statistically significant association between ε4 carrier status and overall ARIA incidence (P = 1.000).

Cognitive and non-cognitive outcomes

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Cognitive outcomes

Cognitive assessments were performed in 45 patients at baseline, decreasing to 29 patients at 3 months and 15 patients at 6 months. MMSE scores were 21.27 ± 4.30 at baseline, 21.66 ± 4.83 at 3 months, and 22.33 ± 5.58 at 6 months, with no statistically significant differences compared to baseline (both P >0.05). Similarly, MoCA scores were 15.90 ± 4.78 at baseline, 16.41 ± 5.38 at 3 months, and 16.38 ± 6.67 at 6 months, showing no significant changes (both P >0.05). CDR-SB scores were 3.16 ± 1.72 at baseline, 3.43 ± 2.17 at 3 months, and 2.30 ± 1.65 at 6 months, with no significant differences from baseline (both P >0.05) [Table 3]. The results of the GLMM analyses based on the multiply imputed datasets are presented in Supplementary Table 2, http://links.lww.com/CM9/C677, and were consistent with the results from the unimputed dataset.

Non-cognitive outcomes

Non-cognitive outcomes, including FAQ and ZBI, were systematically evaluated. No significant changes from baseline were observed in FAQ or ZBI scores at either the 3-month or 6-month follow-ups (all P >0.05) [Table 3]. Results from the GLMM analyses of the multiply imputed datasets can be found in Supplementary Table 2, http://links.lww.com/CM9/C677.

Plasma biomarker outcomes

Plasma biomarker assessments were available for 58 patients at baseline, with follow-up of 53 patients at 3 months and 20 patients at 6 months. Baseline plasma Aβ42 and Aβ40 levels were 4.5 ± 1.2 pg/mL and 81.2 ± 18.7 pg/mL, respectively. After 3 months of lecanemab, levels increased by 7.29% and 16.89%, respectively, and further elevated at 6 months, with significant increases of 21.42% (Aβ42) and 23.53% (Aβ40) from baseline. The mean plasma Aβ42/40 ratio at baseline was 0.06 ± 0.01, which decreased by 9.38% at 3 months (P <0.05) and by 0.34% at 6 months (P >0.05).

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Baseline NfL was 22.4 ± 8.5 pg/mL, with a significant 10.92% increase at 3 months (P <0.05); however, there was a decline to a non-significant 3.76% at 6 months. Plasma GFAP, p-tau181, and p-tau217 levels remained stable over the 6-month treatment period with no significant changes [Supplementary Figure 1, http://links.lww.com/CM9/C677].

MRI outcome

Neuroimaging evaluations included 36 patients at baseline, with 34 patients completing 3-month follow-up and 11 patients completing 6-month scans. In the analysis of zfALFF, which reflects region-specific spontaneous neural activity, only three brain regions – the left inferior frontal gyrus, left cuneus, and right cuneus – showed statistically significant differences after 3 months of lecanemab treatment compared to baseline (adjusted P <0.05), while no significant changes were observed in the remaining regions or at the 6-month follow-up. For zReHo, an index of local functional connectivity synchronization, no significant differences were detected at either time point. Similarly, in the DTI metrics assessing white matter integrity, including axial diffusivity, fractional anisotropy (FA), MD, and radial diffusivity (RD), no significant differences were found in any of the analyzed tract regions at 3- or 6-month post-treatment [Supplementary Table 3, http://links.lww.com/CM9/C678].

PET outcome

Exploratory Aβ-PET analysis was performed in four patients who completed 12 months of lecanemab treatment, with individual trajectories demonstrating reduced amyloid deposition over time. Patient 1 showed a decrease from a baseline SUVR of 1.22 to 1.17 at both 6 and 12 months; Patient 2 declined from 1.38 at baseline to 1.20 at 12 months (6-month data unavailable); Patient 3 decreased from 1.47 at baseline to 1.13 at 6 months and 1.06 at 12 months; and Patient 4 declined from 1.56 at baseline to 1.23 at 6 months and 1.15 at 12 months. All patients exhibited reduced SUVR values at 12

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months relative to baseline. Representative images illustrating the longitudinal Aβ-PET changes following lecanemab treatment are shown in Figure 2. The consistent reduction in Aβ deposition observed in these four patients provides preliminary evidence supporting the effectiveness of lecanemab in clearing cerebral amyloid plaques; however, these findings warrant validation through larger cohort studies with extended follow-up durations.

Discussion

This study addresses the critical gap in lecanemab's clinical use for AD in China. As China′s first real-world evaluation of lecanemab under the "Pioneering and Pilot" policy, it provides initial insights into the safety and preliminary effectiveness of lecanemab in patients with MCI and mild AD. Key findings include a lower incidence of adverse reactions, notably ARIA, cognitive stability after 6 months, and increased plasma Aβ42 and Aβ40 levels. However, the high rate of dropout and the absence of a control group necessitate that these findings, particularly those concerning treatment effectiveness, be interpreted with considerable caution. Despite the short-term follow-up, this study offers valuable preliminary data on lecanemab's safety profile in a Chinese population, which is essential for informing future clinical use and long-term surveillance strategies in China.

Although 56.3% of participants experienced at least one adverse event, no deaths or serious adverse events were reported. The most common adverse events were infusion-related reactions (20.3%), comparable to the Clarity AD trial (24.5%).[12] These reactions were predominantly mild, occurring during the initial infusion and seldom in subsequent administrations. Other frequently reported TEAEs included headaches (9.4%), dizziness (7.8%), decreased appetite (7.8%), and fatigue

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(4.7%), consistent with FAERS data,[25,26] suggesting no significant ethnic differences in the safety profile of lecanemab. Treatment was discontinued due to TEAEs in five participants (7.8%). Specifically, the cases of two patients – one with hydrocephalus and one with atrial fibrillation complicated by heart failure – warrant further discussion regarding a potential association with lecanemab. The atrial fibrillation event occurred in a 79-year-old male with significant cardiovascular comorbidities (diabetes and coronary atherosclerosis) immediately after the 14th infusion. While the patient's history is the most likely cause, the temporal association suggests that lecanemab may have been a precipitating factor. In contrast, the hydrocephalus case in a 57-year-old male lacked imaging or clinical reports for verification. Although ARIA-E – a known side effect of anti-amyloid antibodies – can rarely lead to obstructive hydrocephalus, the lack of confirmatory data in this case makes any causal link to lecanemab highly speculative. Thus, while a contribution of lecanemab cannot be entirely excluded in these cases, the events are more reasonably attributed to underlying conditions or insufficient evidence. Rare adverse events such as sexual dysfunction, head numbness, and somnambulism were also observed, warranting further investigation.

The incidence of ARIA-E (3.1%) and ARIA-H (9.4%) in our cohort was lower than that reported in the Clarity AD trial (ARIA-E: 12.6%; ARIA-H: 16.9%).[27] This difference is unlikely attributable to insufficient follow-up time, as the majority of ARIA events occur within the first 3–6 months of treatment, and all 64 participants in our cohort completed at least the 6-month course. However, longer-term surveillance in larger cohorts is needed to fully characterize the ARIA risk profile in Chinese patients. Instead, the lower ARIA-E incidence may be partly explained by the reduced proportion of APOE ε4 homozygotes in our study (4.7% *vs*. 15.4% in Clarity AD).[12] Similarly, the lower use of antithrombotic agents (7.8% *vs*. 36.5%) may have contributed to the reduced ARIA-H risk,

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given its established association with antithrombotic use.[13] Ethnic differences may also play a role, as suggested by regional variations in other anti-Aβ therapies.[28] It is important to note, however, that our analysis did not identify a statistically significant association between APOE ε4 carrier status and the risk of developing ARIA. This finding should be interpreted with caution, as the statistical power was limited by the small number of APOE ε4 homozygotes (n = 3, 4.7%) in our cohort. The known increased risk of ARIA, particularly ARIA-E, in ε4 homozygotes underscores the need for larger studies to adequately assess genetic influences on lecanemab's safety profile in Chinese populations. Among the eight patients who experienced ARIA-E or ARIA-H, the majority demonstrated favorable clinical outcomes. Treatment discontinuations occurred in two patients: one due to hydrocephalus and another due to the patient's own concerns regarding ARIA. Furthermore, one patient with initial ARIA resolution developed new-onset ARIA-H at 12 months, , highlighting the need for continued vigilance throughout treatment. In summary, while ARIA remains a key safety concern with lecanemab, our findings suggest that with appropriate monitoring, most affected patients can continue treatment safely. Larger prospective studies with longer-term surveillance are needed to definitively characterize the ARIA risk profile in Chinese patients with early AD.

After 6 months of lecanemab treatment, cognitive and functional abilities remained stable, consistent with previous reports of lecanemab's potential to slow disease progression in clinical trials.[29,30] Our Chinese cohort differed notably from the predominantly Western Clarity AD population[6] in several key aspects: (1) younger age (median 68.0 *vs.* 71 years), (2) higher proportion of MCI patients (89.1% *vs*. ~62%), and (3) lower APOE ε4/ε4 homozygosity (4.7% *vs*. 14.8–15.7%). Although these differences may contribute to variations in treatment response, their clinical significance requires verification through extended follow-up. The elevated APOE ε4 rate (67.2%) in

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our cohort, compared to some Chinese studies (15.35–66.7%),[31–33] potentially reflects regional genetic variations or selection bias that warrant investigation in broader population studies. To address potential bias from the high 6-month dropout rate, we conducted sensitivity analyses using GLMM with multiple imputation. The results remained consistent with the unimputed dataset [Supplementary Table 2, http://links.lww.com/CM9/C677], supporting the robustness of our findings. However, a comparison of baseline characteristics indicated that the missingness may not be completely random, as patients lost to follow-up had significantly lower baseline MMSE scores than completers (20.2 ± 4.0 *vs*. 23.3 ± 4.2, $P = 0.025$; Supplementary Table 4, http://links.lww.com/CM9/C677), suggesting that the retained sample might be biased toward individuals with milder cognitive impairment at baseline. Furthermore, to assess whether ApoE ε4 carrier status influenced treatment response, we included an interaction term between ε4 carrier status and time in the GLMM. The analysis revealed no significant interaction effect of ApoE ε4 genotype on the longitudinal changes in cognitive scores (MMSE, MoCA, CDR-SB), functional outcomes (FAQ, ZBI), or plasma biomarker levels (all $P > 0.05$), indicating that the trajectory of treatment response did not differ significantly between ε4 carriers and non-carriers over the 6-month period. However, the small sample size may have limited the statistical power to detect such genetic influences, highlighting the need for larger studies to adequately assess genetic modifiers of lecanemab's efficacy profile in Chinese populations.

Our plasma biomarker findings reveal both consistencies and divergences from the Clarity AD trial.[6] The significant increase in plasma Aβ42 and Aβ40 levels at 6 months (21.42% and 23.53%, respectively) aligns with lecanemab's peripheral sink mechanism, whereby drug-bound Aβ in the bloodstream promotes efflux of Aβ from the brain. This pattern complements the reduced CSF Aβ42/Aβ40 ratio reported in Clarity AD, reflecting a biomarker signature of target engagement that is

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consistent with the proposed mechanism for lecanemab and other anti-Aβ monoclonal antibodies.[6,8,34] In contrast to the marked reductions in plasma p-tau181 and p-tau217 in Clarity AD, these biomarkers remained stable in our cohort. This discrepancy may result from limited statistical power, smaller sample size, attrition, or relatively short treatment duration, which may be insufficient to detect changes in these biomarkers. Therefore, the stability of p-tau should not be interpreted as an absence of disease-modifying effect, but highlights the need for longer-term, larger-scale studies in Chinese populations. Notably, the Aβ42/40 ratio declined initially at 3 months and remained only slightly increased at 6 months, a pattern that contrasts with findings from the Clarity AD trial, where an increase in the Aβ42/40 ratio was reported in supplementary analyses. As lecanemab preferentially targets Aβ42-rich protofibrils[35] but may exert differential effects on Aβ40, further investigation is needed to determine whether this reflects population-specific pharmacology. Collectively, these findings provide preliminary insights into biomarker dynamics among Chinese patients and emphasize the value of further evaluating lecanemab's effects across diverse populations.

The limited neuroimaging changes observed in this study require cautious interpretation. The absence of statistically significant differences in both functional (zfALFF/zReHo) and structural (DTI) metrics does not necessarily indicate lack of treatment effect but may reflect either: (1) inherent limitations in the sensitivity of current neuroimaging biomarkers to detect early amyloid-targeting therapeutic effects, or (2) the relatively short 6-month observation period being insufficient to capture meaningful neural changes. These considerations highlight the need for either more sensitive multimodal imaging protocols or longer-term follow-up to adequately assess amyloid-targeting therapy-related neurobiological changes.

This study has several limitations that should be acknowledged. The primary constraints include

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the small sample size, lack of a control group, and substantial attrition rate. The high dropout rate was largely attributable to unique circumstances surrounding lecanemab's approval in China; following the drug's national authorization and commercial availability during the study period, many participants transitioned to local treatment centers, resulting in administrative discontinuation from our protocol. This attrition considerably diminished the statistical power for longitudinal efficacy assessments, potentially obscuring clinically meaningful treatment effects. Although sensitivity analyses using GLMMs with multiple imputation showed consistent results, the possibility of selection bias persists, as individuals lost to follow-up exhibited significantly lower baseline MMSE scores than completers. These limitations warrant cautious interpretation of the longitudinal efficacy findings.

This study addresses a critical evidence gap regarding the real-world clinical use of lecanemab for AD in China. It provides initial insights into the safety and preliminary effectiveness of lecanemab in patients with MCI or mild AD. Key findings indicate a safety profile, with a lower incidence of adverse reactions, including ARIA. Furthermore, cognitive stability was observed at the 6-month assessment, accompanied by increased plasma levels of Aβ42 and Aβ40. Although these clinical and biomarker outcomes suggest potential therapeutic benefit, the high dropout rate and the absence of a control group necessitate cautious interpretation of effectiveness. Despite the short-term follow-up, this investigation offers valuable preliminary real-world evidence on the safety of lecanemab in a Chinese population, thereby informing current clinical practice and highlighting the need for larger, controlled studies with long-term surveillance.

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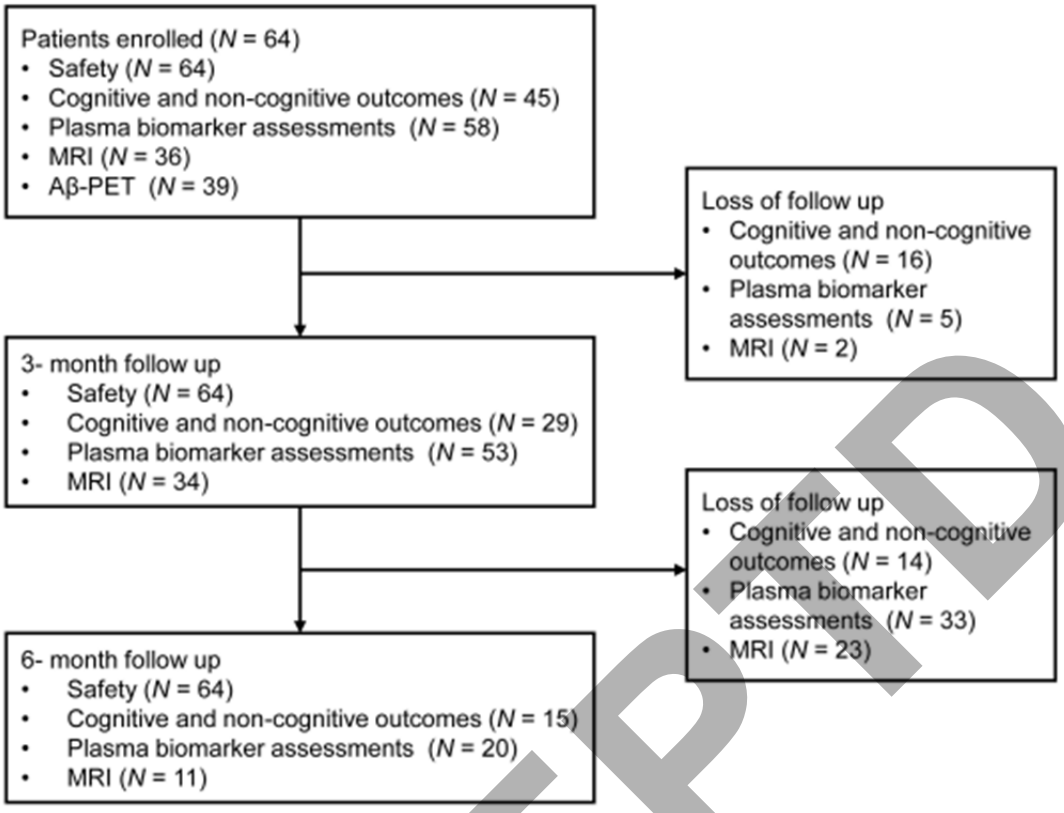

Figure 1: Participant enrollment and follow-up**.** A total of 64 patients were enrolled and underwent assessments at baseline, 3 months, and 6 months. The high dropout rate was primarily attributable to the national approval and commercial availability of lecanemab in China during the study period. Many participants transitioned to receiving treatment at local clinical centers, leading to their administrative discontinuation from the study protocol.

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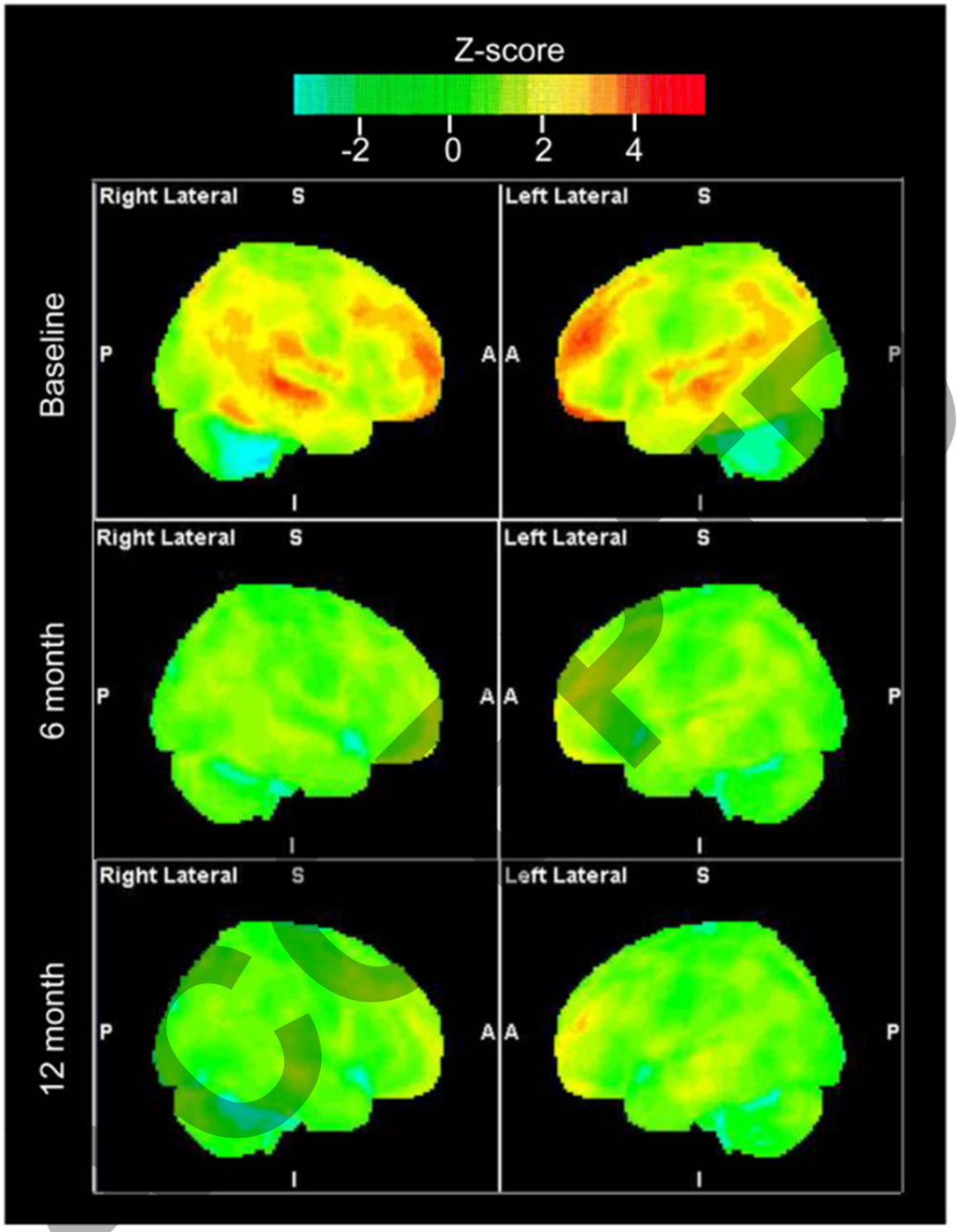

Figure 2: Representative reduction in cerebral amyloid deposition in a patient treated with lecanemab over 12 months. Serial Aβ-PET scans are displayed at baseline (top row) to 6 months (middle row) and 12 months (bottom row) of treatment. For each time point, the right lateral surface view (left column)

and the left lateral surface view (right column) are displayed. Aβ deposition is represented as Z-scores, indicating the number of standard deviations from the mean of a cognitively normal reference cohort. The color scale ranges from –2 to 4, where values near zero (green) reflect normal Aβ levels, and positive values (yellow/red) represent abnormal accumulation. Progressive reduction in Z-scores across time points illustrates amyloid clearance in response to lecanemab. Aβ-PET: Aβ positron emission tomography.

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Table 1: Baseline demographics and clinical characteristics of the early Alzheimer's disease patients treated with lecanemab.

Variables	**Study population (*N* = 64)**		**Variables**	**Study population (*N* = 64)**
Age (years)	68.0 (61.0–		Family history*	18 (28.1)

	74.0)
Sex (male/female)	32/32
Education (years)	12.0 (9.0–15.0) years
BMI (kg/m^2)	22.3 ± 2.9
Disease duration (years)	2.0 (1.0–3.3)
Onset age (years)	64.1 ± 8.3
Clinical stage	
MCI due to AD (global CDR = 0.5)	57 (89.1)
Mild AD (global CDR = 1)	7 (10.9)
Comorbidities	
Stroke	6 (9.4)
Heart disease	2 (3.1)
Hypertension	17 (26.6)
Diabetes	11 (17.2)

Medication use	
Cognition-specific	
Donepezil	44 (68.8)
Memantine	25 (39.1)
Rivastigmine	6 (9.4)
GV-971	14 (21.9)
Huperzine A	2 (3.1)
Combination of cogniton-specific medications	28 (43.8)
Antiplatelet	5 (7.8)
Anticoagulation	0
ApoE genotype	
ε2/ε3	3 (4.7)
ε2/ε4	1 (1.6)

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Hyperlipidemia	10 (15.6)	ε3/ε3	18 (28.1)
Depression and/or anxiety	12 (18.8)	ε3/ε4	39 (60.9)
Head injury	1 (1.6)	ε4/ε4	3 (4.7)

Data are presented as mean ± standard deviation (SD) for normally distributed continuous variables, median (interquartile range, IQR) for non-normally distributed continuous variables, and *n* (%) for categorical variables. *Family history was defined as having at least one first-degree relative (parent, sibling, or children) clinically diagnosed with AD. AD: Alzheimer's disease; ApoE: Apolipoprotein E; BMI: Body mass index; CDR: Clinical dementia rating; IQR: Interquartile range; MCI: Mild cognitive impairment.

Table 2: Lecanemab treatment-emergent adverse events.

Adverse event	**Number (Incidence)**
Total adverse effects	36 (56.3%)
Adverse events lead to discontinuation	5 (7.8%)
Specific TEAEs	
Infusion-related reactions	13 (20.3%)
Flu-like symptoms (fever, chills, generalized aches, joint pain)	6 (9.4%)
Hypotension	3 (4.7%)
Oxygen desaturation	2 (3.1%)
Nausea	1 (1.6%)

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Hypertension	1 (1.6%)
Headache	8 (12.5%)
Dizziness	5 (7.8%)
ARIA-H	6 (9.4%)
Decreased appetite	4 (6.3%)
Fatigue	3 (4.7%)
Bradycardia	3 (4.7%)
Vision changes	2 (3.1%)
Upper respiratory tract infection	2 (3.1%)
Head numbness	2 (3.1%)
ARIA-E	2 (3.1%)
Mucocutaneous hemorrhage	2 (3.1%)
Somnambulism	2 (3.1%)
Dreaminess	1 (1.6%)
Confusion	1 (1.6%)
Diarrhea	1 (1.6%)
Erectile dysfunction	1 (1.6%)
Atrial fibrillation and heart failure	1 (1.6%)
Allergic purpura	1 (1.6%)
Myoclonus	1 (1.6%)

ARIA-E: Amyloid-related imaging abnormalities-edematous; ARIA-H: Amyloid-related imaging abnormalities-hemorrhagic.

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Table 3: Longitudinal cognitive and functional assessments in early Alzheimer's disease patients treated with lecanemab.

	Baseline (*n* = 45)	3 Months (*n* = 29)	6 Months (*n* = 15)	*P* values*	*P* values†
Global cognition					
MMSE	21.27 ± 4.30 (13–30)	21.66 ± 4.83 (14–30)	22.33 ± 5.58 (14–30)	0.199	0.733
MoCA	15.90 ± 4.78 (9–30)	16.41 ± 5.38 (10–30)	16.38 ± 6.67 (10–30)	0.339	0.785
CDR-SB	3.16 ± 1.72 (0.5–8)	3.43 ± 2.17 (0–10)	2.30 ± 1.65 (0.5–6)	0.391	0.357
Functional outcomes					
FAQ	6.33 ± 5.44 (0–20)	7.34 ± 6.33 (0–21)	6.50 ± 6.16 (0–17)	0.441	0.632
Caregiver burden					
ZBI	16.10 ± 9.58	15.96 ±	14.71 ± 7.57	0.210	0.598

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	(3–42)	13.16 (0–59)	(5–24)

The mean changes, standard deviations, and *P* values adjusted relative to baseline were calculated using a generalized mixed-effects model. Fixed effects included follow-up time, age, gender, educational level, ApoE4 carrier status, while individual patient effects were treated as random effects, with the baseline value included as a covariate. *Comparison between 3 months and baseline. †Comparison between 6 months and baseline. Data are reported as mean ± standard deviation (SD) and range. CDR-SB: Clinical dementia rating–sum of boxes; FAQ: Functional activities questionnaire; MMSE: Mini-mental state examination; MoCA: Montreal Cognitive Assessment; ZBI: Zarit Burden Interview.

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